The Great Recession: Understanding the causes

Paul Willen

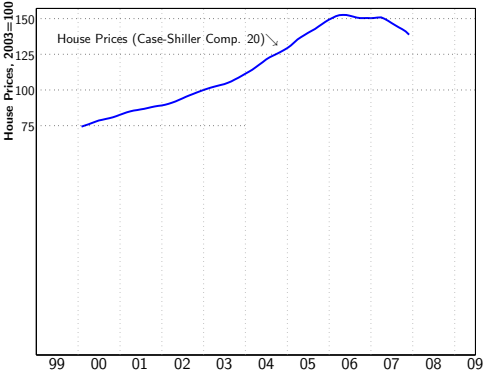
Federal Reserve Bank of Boston, Harvard and NBER

2410, March 9, 2017

These notes reflect the views of the author and don't necessarily reflect the official positions of the Federal Reserve Bank of Boston or the Federal Reserve System.
Prepared for ec2410 at Harvard, Spring 2017.
©2017 by Paul S. Willen.

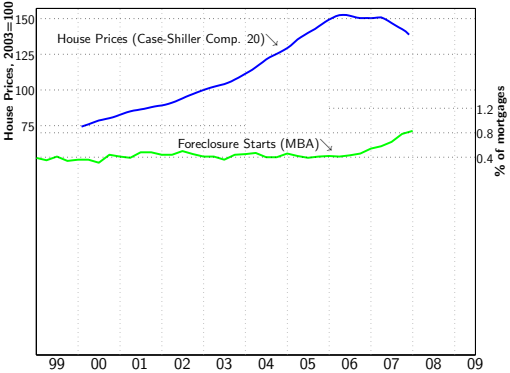
2000s business cycle and housing

- Boom and bust in housing at heart of crisis



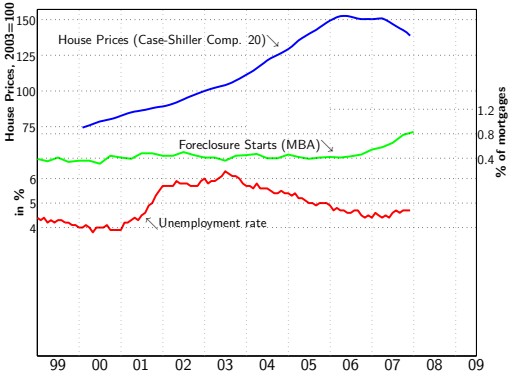
2000s business cycle and housing

- Boom and bust in housing at heart of crisis



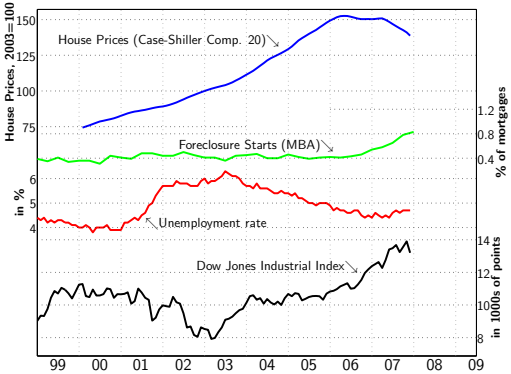
2000s business cycle and housing

- Boom and bust in housing at heart of crisis



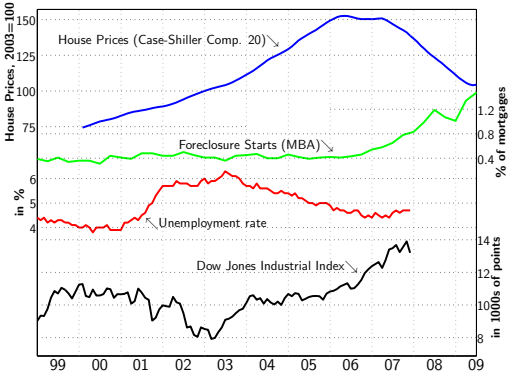
2000s business cycle and housing

- Boom and bust in housing at heart of crisis

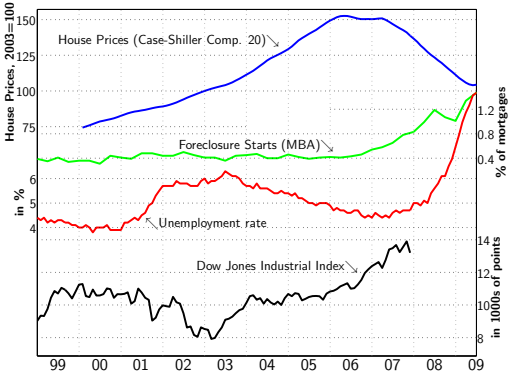


2000s business cycle and housing

- Boom and bust in housing at heart of crisis

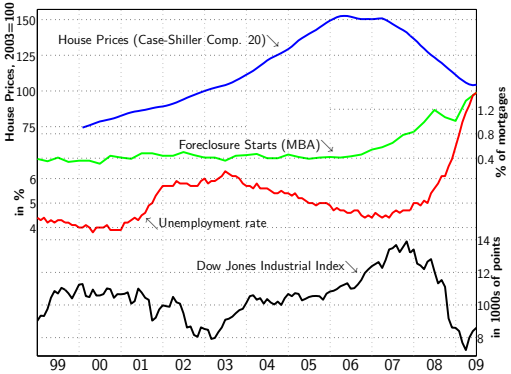


2000s business cycle and housing



- Boom and bust in housing at heart of crisis
- Focus on the boom in lending

2000s business cycle and housing



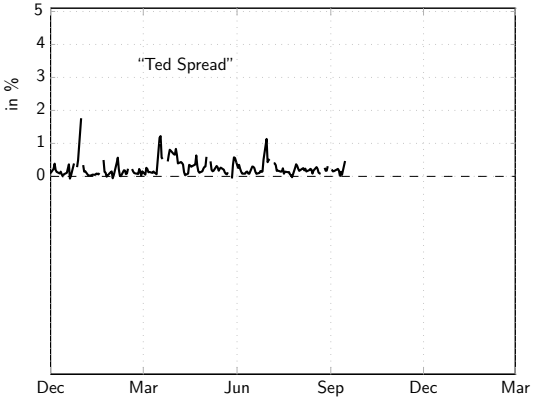
- Boom and bust in housing at heart of crisis
- Focus on the boom in lending
- and the connection to the foreclosures.

The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild

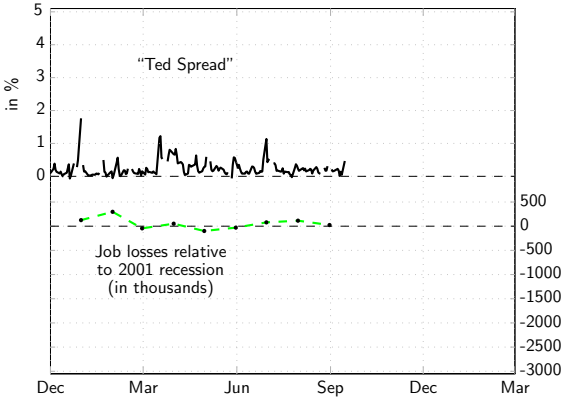
The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild



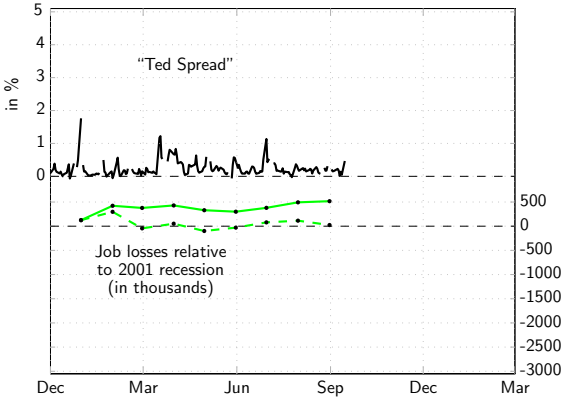
The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild



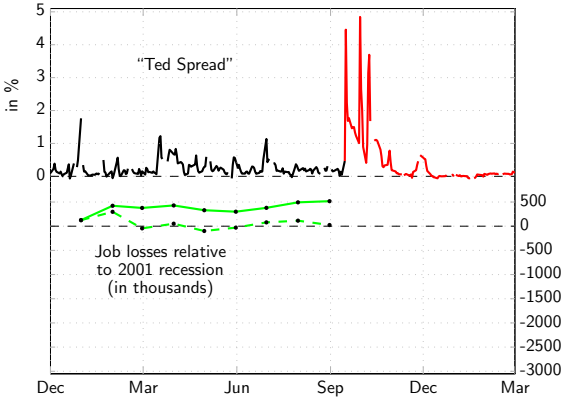
The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild
- Lehman fails on September 15, 2008
  - Worst job losses since Great Depression



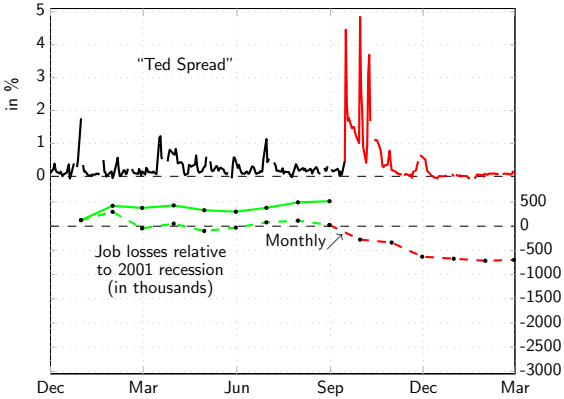
The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild
- Lehman fails on September 15, 2008
  - Worst job losses since Great Depression



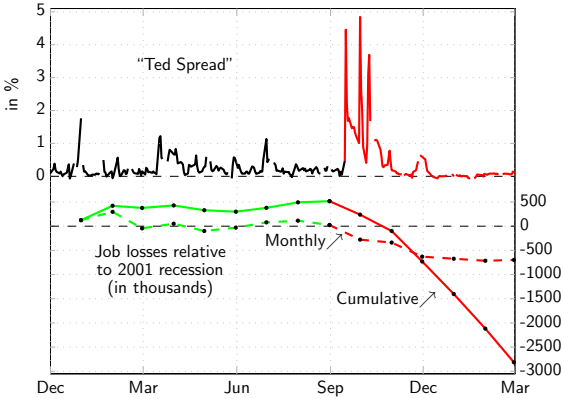
The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild
- Lehman fails on September 15, 2008
  - Worst job losses since Great Depression



The financial crisis

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild
- Lehman fails on September 15, 2008
  - Worst job losses since Great Depression

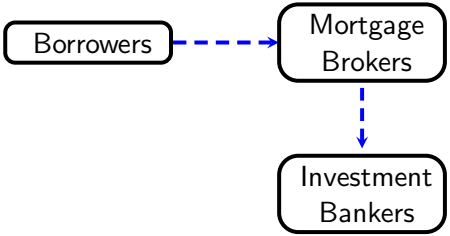


Why did people make so many bad decisions

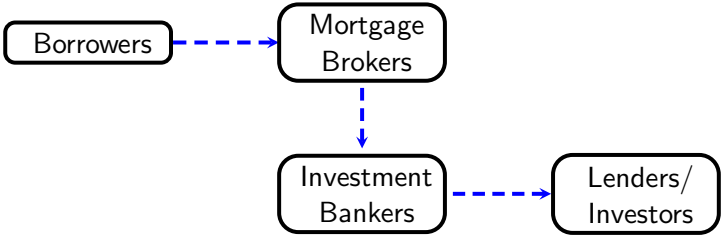
Borrowers

Why did people make so many bad decisions

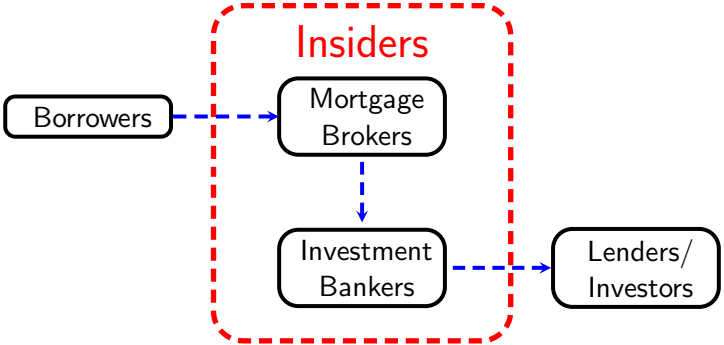
Why did people make so many bad decisions



Why did people make so many bad decisions

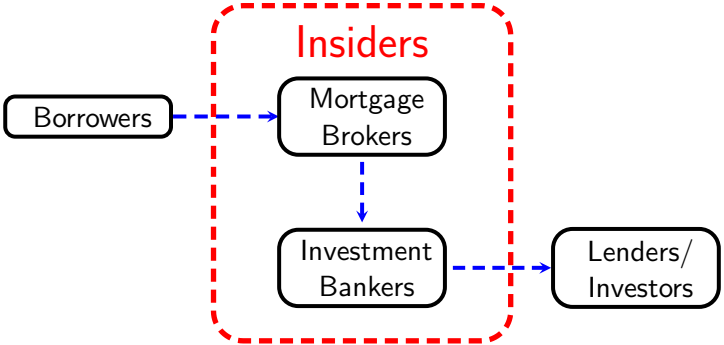


Why did people make so many bad decisions



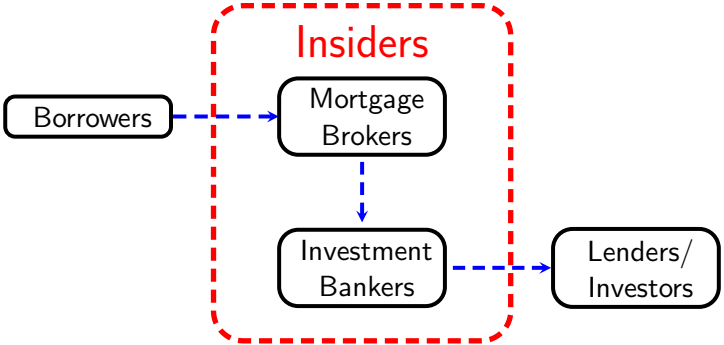
- The insiders were the problem.

Why did people make so many bad decisions



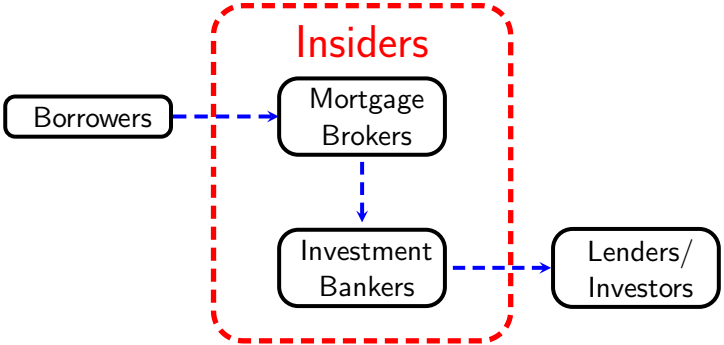
- The insiders were the problem.
- Deceive investors and borrowers.

Why did people make so many bad decisions



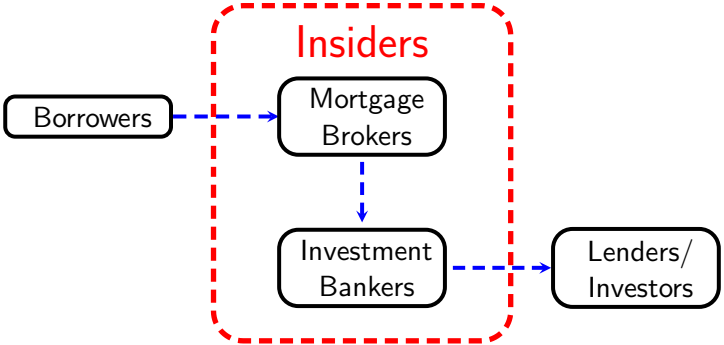
- The insiders were the problem.
- Deceive investors and borrowers.
- Brokers push "exploding" mortgages.

Why did people make so many bad decisions



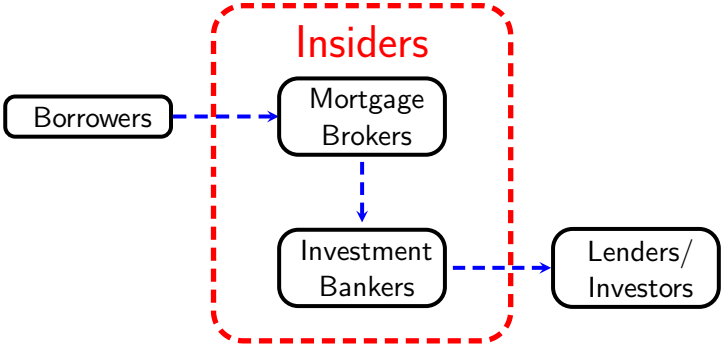
- The insiders were the problem.
- Deceive investors and borrowers.
- Brokers push "exploding" mortgages.
- Bankers create "toxic" securities

Why did so many people make bad decisions?



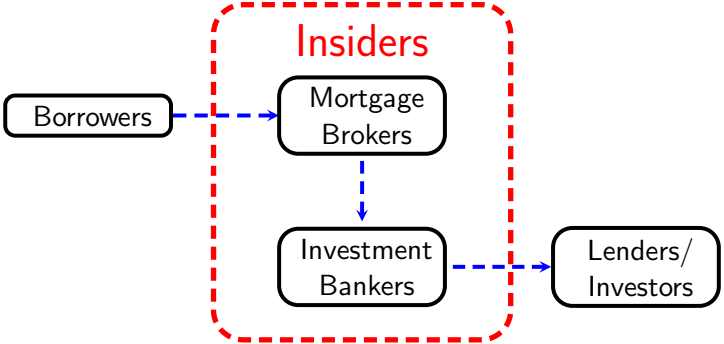
- Show 12 facts that contradict the story

Why did so many people make bad decisions?



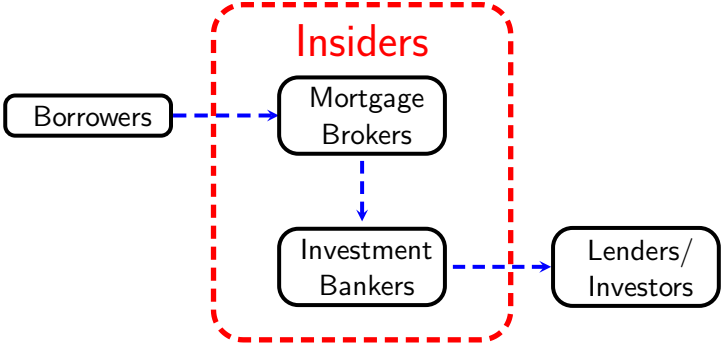
- Show 12 facts that contradict the story
- Exploding mortgages not a problem

Why did so many people make bad decisions?



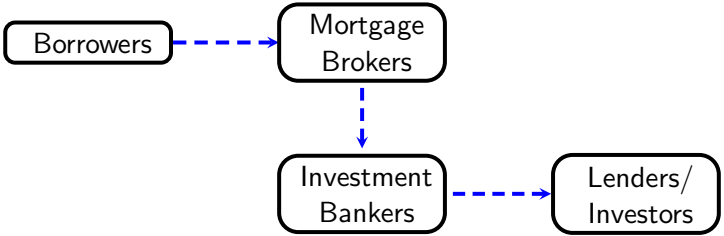
- Show 12 facts that contradict the story
- Exploding mortgages not a problem
- Banks did not hide what was going on.

Why did so many people make bad decisions?



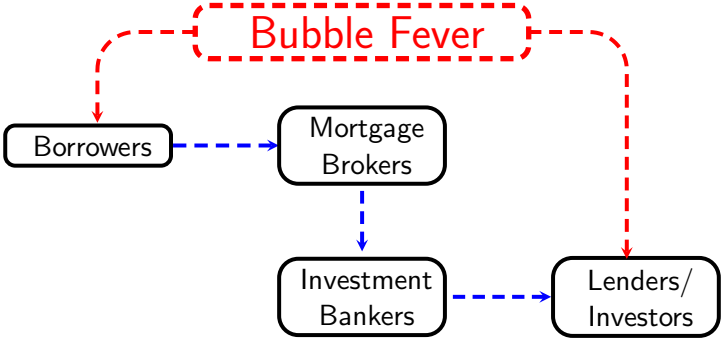
- Show 12 facts that contradict the story
- Exploding mortgages not a problem
- Banks did not hide what was going on.
- Investors could and did figure it out possible

Why did so many people make bad decisions?



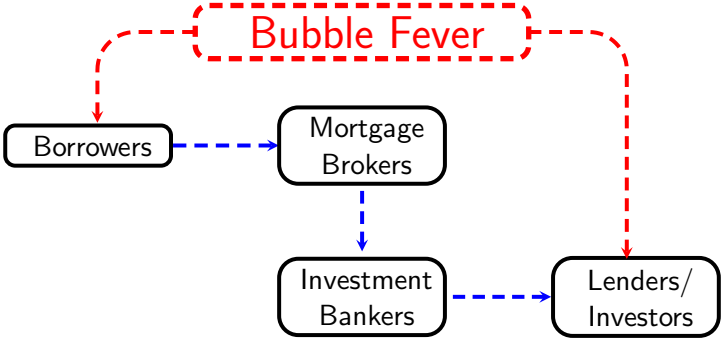
- So what was the problem?

Why did so many people make bad decisions?



- So what was the problem?
- Bubble Fever: Borrowers and lenders were trying to cash in on the biggest real estate boom in American history

Why did so many people make bad decisions?



- So what was the problem?
- Bubble Fever: Borrowers and lenders were trying to cash in on the biggest real estate boom in American history
- It is hard to stop consenting adults.

Role of Adjustable Rate Mortgages

Role of Adjustable Rate Mortgages

- Adjustable Rate Mortgages.

Role of Adjustable Rate Mortgages

- Adjustable Rate Mortgages.
- Payment Shocks
- Large national sample of loans on which lender initiated foreclosure.
  - Consistent with other datasets

Role of Adjustable Rate Mortgages

- Adjustable Rate Mortgages.
- Payment Shocks
- Large national sample of loans on which lender initiated foreclosure.
  - Consistent with other datasets

		2007	2008	2009	2010	All
Prior to delinquency spell that led to foreclosure...						
% of loans with...	Payment increase	12%	17%	11%	9%	12%
	Payment reduction	0%	0%	4%	8%	4%
	No change since orig.	88%	82%	85%	83%	84%
FRM share		38%	48%	62%	74%	59%
ARMs prior to reset		44%	32%	20%	15%	24%
ARMs that reset but payment stayed the same or fell		6%	2%	7%	2%	5%
# obs in thous.		374	641	874	756	2,646

Fact 2: No loan was designed to fail

- Some loans were *prima facie* evidence of predatory lending.

Fact 2: No loan was designed to fail

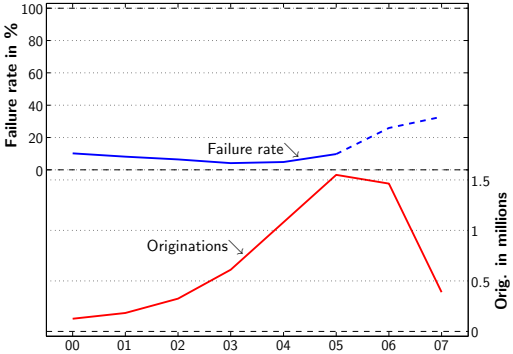
- Some loans were *prima facie* evidence of predatory lending.
- No reasonable person would ever get a no-doc loan.

Fact 2: No loan was designed to fail

- Some loans were *prima facie* evidence of predatory lending.
- No reasonable person would ever get a no-doc loan.
- "Designed to fail"

Fact 2: No loan was designed to fail

- Some loans were *prima facie* evidence of predatory lending.
- No reasonable person would ever get a no-doc loan.
- "Designed to fail"



Fact 3: Little innovations

- Low Documentation Mortage, January 9, 1985.

Fact 3: Little innovations

- Low Documentation Mortage, January 9, 1985.

THE BEST MORTGAGE DEAL IN TOWN - JUST GOT BETTER!

Introducing Free Gifts from GELT Funding

GELT FUNDING, the company that introduced "hassle free" mortgages, now makes getting a mortgage not only easier but downright enjoyable. We will give you more than just a great rate on your mortgage, but also a FREE gift along with it!

This is our way of saying thank you, for making us one of the most popular mortgage bankers in town.

We offer a full range of mortgages at the absolute lowest rates.

- First & Second Mortgages
- Fixed & Adjustable Mortgages (with life time cap)
- Mortgages for Condos & Co-ops
- Refinancing & Assumable
- No Income Verification

FREE GIFT SELECTION

$50,000 - $99,999 Mortgage
Cannon A 1 w/1.4 lens
GE or Litton Microwave Oven
GE or Crown Dishwasher

$100,000 - $149,999 Mortgage
MC or GE Video cassette Recorder
Jenn-Air or GE 25" Color TV
Electrolux Vacuum Cleaner

$150,000 and over
GE 20 cubic ft. frost free Refrigerator
GE Stack Washer & Dryer
Onkyo Model #228 Stereo System

Mortgage Banker

GELT FUNDING
718-854-1700

Offer limited to new applicants. Time must close by April X, 1985.
Offer may be withdrawn without notice

Fact 4: Low down payments were not new

[I]n just a few short years, all of the venerable rules governing the relationship between borrower and lender went out the window, starting with the elimination of the requirements that a borrower put down a substantial amount of cash in a property. (Morgenson and Rosner, 2011)

Fact 4: Low down payments were not new

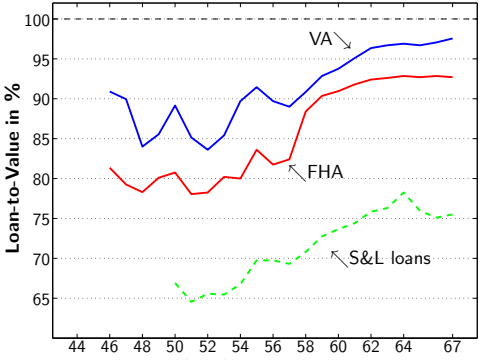
[I]n just a few short years, all of the venerable rules governing the relationship between borrower and lender went out the window, starting with the elimination of the requirements that a borrower put down a substantial amount of cash in a property. (Morgenson and Rosner, 2011)

- No downpayment loans had been around since the GI Bill.

Fact 4: Low down payments were not new

[I]n just a few short years, all of the venerable rules governing the relationship between borrower and lender went out the window, starting with the elimination of the requirements that a borrower put down a substantial amount of cash in a property. (Morgenson and Rosner, 2011)

- No downpayment loans had been around since the GI Bill.



Securitization was not new

- Securitization was not new

Securitization was not new

- Securitization was not new
- Big shift.

Securitization was not new

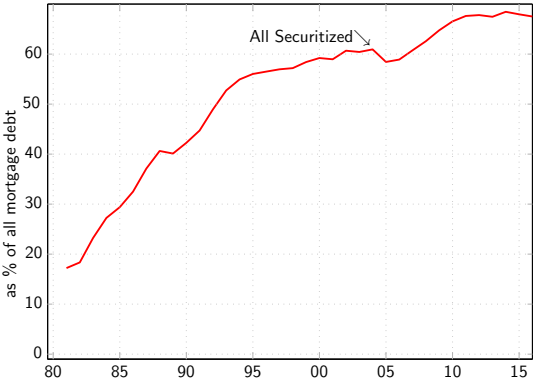
- Securitization was not new
- Big shift.
- Long before the boom.

Securitization was not new

- Securitization was not new
- Big shift.
- Long before the boom.
- "Mortgage Companies": stand-alone companies that originate but don't hold mortgages.

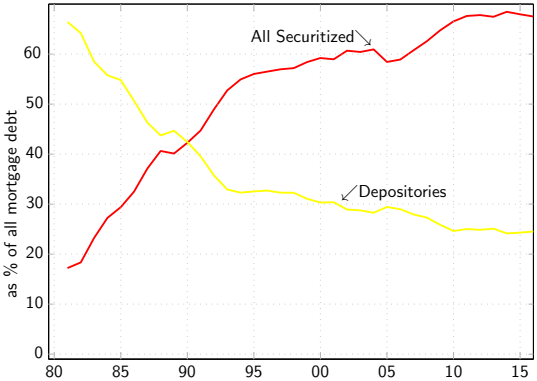
Securitization was not new

- Securitization was not new
- Big shift.
- Long before the boom.
- "Mortgage Companies": stand-alone companies that originate but don't hold mortgages.
  - The Mortgage Bankers Association was founded in 1909.



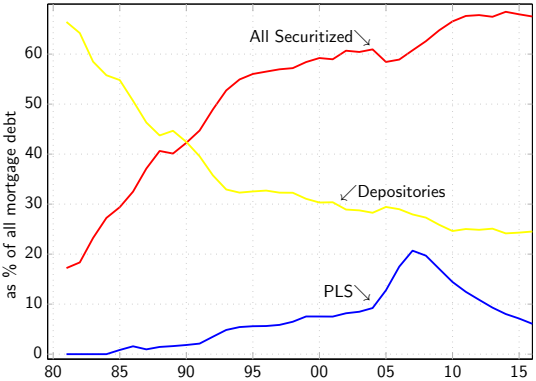
Securitization was not new

- Securitization was not new
- Big shift.
- Long before the boom.
- "Mortgage Companies": stand-alone companies that originate but don't hold mortgages.
  - The Mortgage Bankers Association was founded in 1909.
  - As far back as the 1950s, MCs accounted for 25 percent of new originations



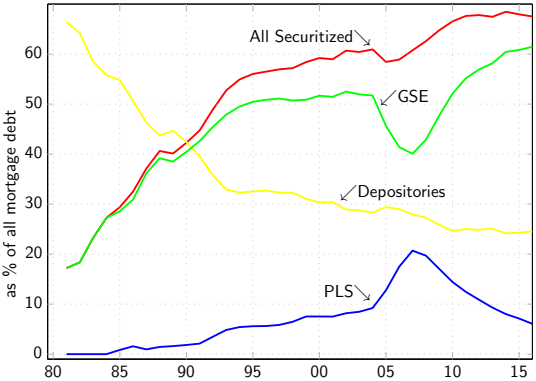
Securitization was not new

- Securitization was not new
- Big shift.
- Long before the boom.
- "Mortgage Companies": stand-alone companies that originate but don't hold mortgages.
  - The Mortgage Bankers Association was founded in 1909.
  - As far back as the 1950s, MCs accounted for 25 percent of new originations

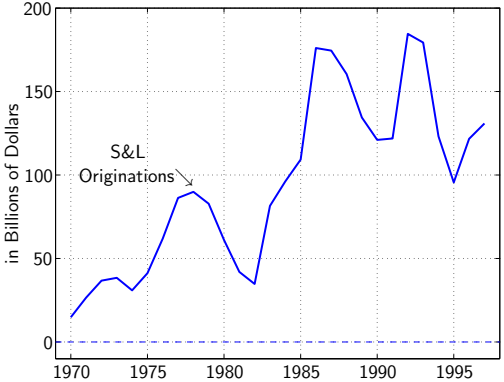


Securitization was not new

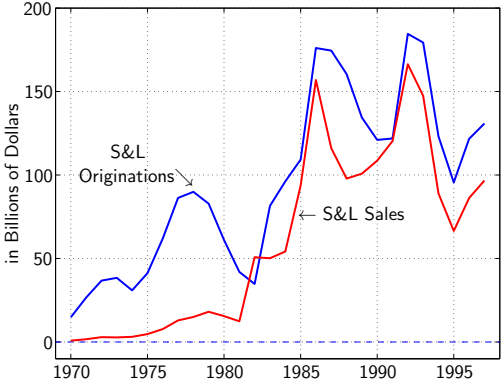
- Securitization was not new
- Big shift.
- Long before the boom.
- "Mortgage Companies": stand-alone companies that originate but don't hold mortgages.
  - The Mortgage Bankers Association was founded in 1909.
  - As far back as the 1950s, MCs accounted for 25 percent of new originations



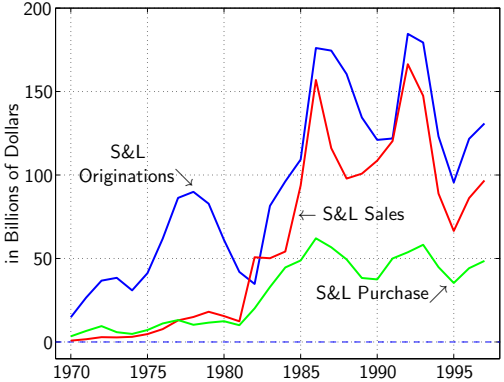
Savings and Loans



Savings and Loans



Savings and Loans



The central role of securitization

	Outcome			
	1	2	3	X
Homeowner A	100	80	100	80
Homeowner B	100	100	70	70
Pool income	200	180	170	150
"AAA" rated security	160	160	160	150
"BBB" rated security	20	20	10	0
Equity	20	0	0	0
Probability	30	50	20	0

- Why would investors buy B and equity?

The central role of securitization

	Outcome			
	1	2	3	X
Homeowner A	100	80	100	80
Homeowner B	100	100	70	70
Pool income	200	180	170	150
"AAA" rated security	160	160	160	150
"BBB" rated security	20	20	10	0
Equity	20	0	0	0
Probability	30	50	20	0

- Why would investors buy B and equity?
- Did they not know that there was a state X?

The central role of securitization

	Outcome			
	1	2	3	X
Homeowner A	100	80	100	80
Homeowner B	100	100	70	70
Pool income	200	180	170	150
"AAA" rated security	160	160	160	150
"BBB" rated security	20	20	10	0
Equity	20	0	0	0
Probability	30	50	20	0

- Why would investors buy B and equity?
- Did they not know that there was a state X?
- Did they not know how bad the loans were?

The central role of securitization

	Outcome			
	1	2	3	X
Homeowner A	100	80	100	80
Homeowner B	100	100	70	70
Pool income	200	180	170	150
"AAA" rated security	160	160	160	150
"BBB" rated security	20	20	10	0
Equity	20	0	0	0
Probability	30	50	20	0

- Why would investors buy B and equity?
- Did they not know that there was a state X?
- Did they not know how bad the loans were?
- Did they not understand the moral hazard problem?

Fact 7: Investors had lots of information

- What did investors know?
- A lot.
- At origination: http://www.sec.gov/Archives/edgar/data/1375560/000119312506194735/dfwp.htm
- After origination http://www.ctslink.com/

List of Securities in Abacus Deal

Reference Portfolio

Security	Type	Notional Amount	CUSIP	Fitch	Moody's	S&P	Base WAL (yrs)	Dated Date	Legal Final	Servicer
ABFC 2006-OPT1 M8	Subprime	22,222,222	000750AM4	BBB	Baa2	BBB	3.9	8/10/2006	9/25/2036	OCMC
ABFC 2006-OPT2 M8	Subprime	22,222,222	000750AP2	BBB	Baa2	BBB	4.1	10/12/2006	10/25/2036	OCMC
ABSHE 2006-HE3 M7	Subprime	22,222,222	04541GXK3	BBB	Baa2	BBB	3.8	4/17/2006	3/25/2036	OCMC
ABSHE 2006-HE4 M7	Subprime	22,222,222	04544GAP4	BBB	Baa2	BBB	3.8	4/28/2006	5/25/2036	SPS
ACE 2006-FM2 M8	Midprime	22,222,222	00442CAN9		Baa2	BBB	4.5	10/30/2006	8/25/2036	WFB
ACE 2006-OP2 M9	Subprime	22,222,222	00441YAP7		Baa2	BBB-	4.3	10/30/2006	8/25/2036	WFB
ARSI 2006-W1 M8	Subprime	22,222,222	040104RQ6	BBB+	Baa2	BBB+	3.8	2/7/2006	3/25/2036	AQMC
CARR 2006-FRE1 M8	Subprime	22,222,222	14430BAN5	BBB+	Baa2	A	3.8	6/28/2006	7/25/2036	FREM
CARR 2006-FRE2 M8	Subprime	22,222,222	14454AAN8		Baa2	BBB+	4.2	10/18/2006	10/25/2036	FREM
CARR 2006-NC1 M8	Midprime	22,222,222	14453TFF2	BBB	Baa2	BBB+	3.6	2/8/2006	1/25/2036	NCMC
CARR 2006-NC2 M8	Subprime	22,222,222	14454FAA1	BBB	Baa2	BBB	3.8	6/21/2006	6/25/2036	CARR
CARR 2006-NC3 M9	Subprime	22,222,222	14452BAN6	BBB-	Baa2	BBB-	4.0	8/10/2006	8/25/2036	NCMC
CARR 2006-OPT1 M8	Subprime	22,222,222	14453TFY7	BBB+	Baa2	A-	3.6	3/14/2006	2/25/2036	OCMC
CMLTI 2006-AMC1 M8	Subprime	22,222,222	17309PAL0		Baa2	BBB	4.1	9/28/2006	9/25/2036	AQMC
CMLTI 2006-NC1 M8	Subprime	22,222,222	172983AN8		Baa2	BBB	3.8	6/29/2006	8/25/2036	WFB
CMLTI 2006-WFH2 M9	Midprime	22,222,222	17309MAN3		Baa2	BBB-	4.0	8/30/2006	8/25/2036	WFB
CMLTI 2006-WMC1 M8	Midprime	22,222,222	17307GZF4	A-	Baa2	BBB+	3.7	1/31/2006	12/25/2035	WFB
CMLTI 2007-WFH1 M8	Subprime	22,222,222	17311CAM3		Baa2	BBB-	4.5	2/9/2007	1/25/2037	WFB
CWL 2006-24 M8	Subprime	22,222,222	232439AN1		Baa2	BBB	4.9	12/29/2006	5/25/2037	CHLS
FFML 2006-FF11 M8	Midprime	22,222,222	32028PAP0	BBB	Baa2	BBB	3.9	9/6/2006	8/25/2036	WFB
FFML 2006-FF12 M8	Midprime	22,222,222	32027GAN6	BBB	Baa2	BBB	4.2	8/25/2006	9/25/2036	ALS
FFML 2006-FF14 M8	Midprime	22,222,222	32027LAP0	BBB	Baa2	BBB	4.2	9/25/2006	10/25/2036	AURA
FFML 2006-FF15 M8	Midprime	22,222,222	32028GAP0	BBB	Baa2	BBB	4.3	10/25/2006	11/25/2036	AURA
FFML 2006-FF16 M8	Midprime	22,222,222	32027KAN0		Baa2	BBB+	4.3	11/30/2006	12/25/2036	NCHL
FFML 2006-FF17 M8	Midprime	22,222,222	32028KAP1	BBB	Baa2	BBB	4.4	11/25/2006	12/25/2036	ALS
FFML 2006-FF7 M8	Midprime	22,222,222	32027FAP1	BBB	Baa2	BBB	3.6	5/31/2006	5/25/2036	WFB
FFML 2006-FF9 M8	Midprime	22,222,222	32027SAP3	BBB+	Baa2	BBB+	3.7	7/7/2006	6/25/2036	WFB
FHLT 2006-A M7	Subprime	22,222,222	35729RAN6	BBB+	Baa2	BBB	3.9	5/10/2006	5/25/2036	WFB
FHLT 2006-B M8	Midprime	22,222,222	35729GAN8	BBB+	Baa2	BBB	4.4	8/3/2006	8/25/2036	WFB
FMIC 2006-2 M8	Midprime	22,222,222	31659EAM0		Baa2	BBB+	4.1	7/6/2006	7/25/2036	WFB
FMIC 2006-3 M8	Midprime	22,222,222	31659BAN0		Baa2	BBB	4.4	10/27/2006	11/25/2036	WFB
GSAMP 2006-FM2 M8	Midprime	22,222,222	36245DAN0		Baa2	BBB+	4.0	9/29/2006	9/25/2036	WFB
HEAT 2006-3 M8	Midprime	22,222,222	43708AUZ7	BBB+	Baa2	BBB+	3.5	3/30/2006	7/25/2036	SPS
HEAT 2006-5 M8	Midprime	22,222,222	43709BAQ3	BBB+	Baa2	BBB+	3.8	6/25/2006	10/25/2036	WFB
HEAT 2006-6 M8	Midprime	22,222,222	43709TAP3	A-	Baa2	A-	4.0	8/1/2006	11/25/2036	SPS
HEAT 2006-7 M8	Midprime	22,222,222	43709NAP8	BBB+	Baa2	BBB+	4.2	10/3/2006	1/25/2037	SPS
HEAT 2006-8 M8	Midprime	22,222,222	43709QAP1	BBB	Baa2	BBB+	4.4	12/1/2006	3/25/2037	SPS
IWIS 2006-HE3 B2	Midprime	22,222,222	466024UM6	BBB	Baa2	BBB	4.8	9/29/2006	12/25/2037	WFB
JPMAC 2006-CW2 MV8	Midprime	22,222,222	46629BBA6	BBB	Baa2	BBB	4.3	8/8/2006	8/25/2036	CWHL
JPMAC 2006-FRE1 M8	Midprime	22,222,222	4662KLFY7	BBB	Baa2	BBB	3.6	1/27/2006	5/25/2035	JPM
JPMAC 2006-WMC3 M8	Midprime	22,222,222	46629KAP4	BBB	Baa2	BBB	4.3	9/14/2006	8/25/2036	JPM
LBMLT 2006-11 M8	Midprime	22,222,222	54251UNB8		Baa2	BBB	4.7	12/14/2006	12/25/2036	WMB
LBMLT 2006-4 M8	Midprime	22,222,222	54251MAN4			A-	3.9	5/9/2006	5/25/2036	WMB
LBMLT 2006-6 M8	Midprime	22,222,222	54251RAN3	BBB+	Baa2	BBB+	4.2	7/26/2006	7/25/2036	WMB
LBMLT 2006-7 M8	Midprime	22,222,222	54251TAN9	BBB+	Baa2	A-	4.2	8/31/2006	8/25/2036	WMB

As of February 26, 2007. Goldman Sachs neither represents nor provides any assurances that the actual Reference Portfolio on the Closing Date or any future date will have the same characteristics as represented above. See the final Offering Circular for the Initial Reference Portfolio.

Reference Obligations are designated as "Midprime" herein if the weighted average FICO score of the underlying collateral that secures such Reference Obligation is greater than 625. All other Reference Obligations are designated as "Subprime" herein.

55

List of Securities in Abacus Deal

Goldman Sachs

Reference Portfolio

Security	Type	Notional Amount	CUSIP	Fitch	Moody's	S&P	Base WAL (yrs)	Dated Date	Legal Final	Servicer
LBMLT 2006-WL1 M8	Midprime	22,222,222	542514RD8		Baa2	BBB	3.1	2/8/2006	1/25/2036	LBMC
MABS 2006-HE3 M9	Subprime	22,222,222	576455AN9		Baa2	BBB-	4.5	12/28/2006	11/25/2036	WFB
MABS 2006-NC2 M9	Subprime	22,222,222	552758AP2	BBB	Baa2	BBB-	4.2	9/28/2006	8/25/2036	WFB
MABS 2006-WMC4 M8	Midprime	22,222,222	576455AP7		Baa2	BBB+	4.6	11/30/2006	10/25/2036	WFB
MLMI 2006-WMC1 B2A	Midprime	22,222,222	59020U4H5		Baa2	BBB+	3.6	2/14/2006	1/25/2037	WCC
MSAC 2006-HE7 B2	Subprime	22,222,222	617556MAP9		Baa2	BBB	4.9	10/31/2006	9/25/2036	CWHL
MSAC 2006-HE8 B2	Midprime	22,222,222	617555AP7		Baa2	BBB	5.1	11/29/2006	10/25/2036	WFB
MSAC 2006-NC4 B2	Subprime	22,222,222	61746LAN2	BBB	Baa2	BBB	4.5	6/23/2006	6/25/2036	WFB
MSAC 2006-NC5 B3	Midprime	22,222,222	61748BAQ8		Baa2	BBB-	5.3	11/28/2006	10/25/2036	CWHL
MSAC 2006-WMC1 B2	Midprime	22,222,222	61744CXV3	BBB+	Baa2	A-	4.2	1/26/2006	12/25/2035	JPM
MSAC 2006-WMC2 B2	Midprime	22,222,222	61748AKF9	BBB	Baa2	BBB	4.7	6/28/2006	7/25/2036	WFB
MSAC 2007-NC1 B2	Subprime	22,222,222	617505AN2		Baa2	BBB	5.3	1/26/2007	11/25/2036	CWHL
MSC 2006-HE2 B2	Midprime	22,222,222	61745YFD8	BBB	Baa2	BBB+	4.5	4/28/2006	3/25/2036	WFB
MSIX 2006-2 B2	Midprime	22,222,222	617463AM6		Baa2	BBB	5.0	11/28/2006	11/25/2036	SAX
NHEL 2006-5 M8	Subprime	22,222,222	65889YAN2		Baa2	BBB+	4.0	9/28/2006	11/25/2036	NOVA
NHEL1 2006-FM1 M8	Midprime	22,222,222	65539HCF3		Baa2	BBB+	3.3	1/30/2006	11/25/2035	WFB
NHEL1 2006-FM2 M8	Midprime	22,222,222	65537FAN1	BBB+	Baa2	BBB+	4.1	10/31/2006	7/25/2036	WFB
NHEL1 2006-HE3 M8	Subprime	22,222,222	65538QAN8	BBB+	Baa2	BBB+	4.0	8/31/2006	7/25/2036	WFB
OOMLT 2007-1 M8	Subprime	22,222,222	68400DAP9		Baa2	BBB	4.3	1/24/2007	1/25/2037	OOMC
SABR 2006-FR1 B2	Midprime	22,222,222	813759VJY3	BBB+	Baa2	A-	4.6	2/23/2006	11/25/2035	HSC
SABR 2006-FR3 B2	Subprime	22,222,222	813765AH7	BBB+	Baa2	BBB	5.0	8/3/2006	5/25/2036	HSC
SABR 2006-HE2 B2	Subprime	22,222,222	81377AAAM	BBB+	Baa2	BBB	4.1	9/28/2006	7/25/2036	HSC
SAIL 2006-4 M7	Subprime	22,222,222	86360WAM4	BBB	Baa2	BBB	4.1	6/25/2006	7/25/2036	ALS
SASC 2006-EQ1A M8	Subprime	22,222,222	86360RAN3		Baa2	BBB	5.2	7/17/2006	5/25/2036	AURA
SASC 2006-OPT1 M7	Subprime	22,222,222	86359UAN9	BBB	Baa2	BBB	3.7	4/25/2006	4/25/2036	AURA
SURF 2007-BC1 B2	Subprime	22,222,222	84752BAQ2		Baa2	BBB	4.9	1/24/2007	1/25/2038	WCC
SVHE 2006-EQ2 M8	Midprime	22,222,222	83611XAM6	BBB	Baa2	BBB	4.6	12/28/2006	1/25/2037	OLS
SVHE 2006-OPT1 M7	Subprime	22,222,222	83611MMF2	BBB+	Baa2	BBB	3.6	3/10/2006	3/25/2036	OOMC
SVHE 2006-OPT2 M7	Subprime	22,222,222	83611MMT2		Baa2	A-	3.6	4/7/2006	5/25/2036	OOMC
SVHE 2006-OPT3 M7	Subprime	22,222,222	83611MPR3		Baa2	BBB	3.7	5/12/2006	6/25/2036	OOMC
SVHE 2006-OPT5 M8	Subprime	22,222,222	83612CAN9		Baa2	BBB	4.2	6/19/2006	7/25/2036	OOMC
ABSHE 2006-HE7 M9	Subprime	22,222,222	04544QAP2	BBB-	Baa2	BBB-	4.4	11/30/2006	11/25/2036	SPS
BSABS 2006-HE9 M9	Subprime	22,222,222	07388MAP2		Baa2	BBB-	4.4	11/30/2006	11/25/2036	EMC
CMLTI 2007-AMC1 M8	Subprime	22,222,222	17311BAL7		Baa2	BBB	4.6	3/9/2007	12/25/2036	CWHL
FFML 2007-FF1 B2	Midprime	22,222,222	32028TAN7		Baa2	BBB	4.8	1/26/2007	1/25/2038	HLS
HASC 2006-HE2 M8	Midprime	22,222,222	44328BAP3	BBB+	Baa2	BBB+	4.3	12/5/2006	12/25/2036	CMB
HEAT 2007-1 M8	Midprime	22,222,222	43719LAN4	BBB	Baa2	BBB+	4.5	2/1/2007	5/25/2037	SPS
LBMLT 2006-8 M8	Midprime	22,222,222	542514UN6		Baa2	A-	4.4	9/21/2006	9/25/2036	WMB
LBMLT 2006-9 M8	Midprime	22,222,222	542514WAN2		Baa2	BBB+	4.4	10/12/2006	10/25/2036	WMB
MLMI 2006-HE B3	Subprime	22,222,222	59022XAN6		Baa2	BBB-	4.6	12/28/2006	11/25/2037	WCC
MLMI 2006-OPT1 B2	Subprime	22,222,222	59022VAN1		Baa2	BBB	3.9	9/26/2006	8/25/2037	OOMC
MSAC 2007-HE1 B2	Subprime	22,222,222	617524AP3		Baa2	BBB	5.2	1/26/2007	11/25/2036	SM
OOMLT 2006-3 M9	Subprime	22,222,222	683688AM5		Baa2	BBB-	4.0	10/27/2006	2/25/2037	OOMC
SASC 2006-WF3 M9	Subprime	22,222,222	86361EAP6	BBB-	Baa2	BBB-	4.3	9/25/2006	9/25/2036	ALS
SVHE 2006-OPT4 M7	Subprime	22,222,222	83611YAM4		Baa2	BBB+	3.6	5/26/2006	6/25/2036	OOMC

ACA Capital

As of February 26, 2007. Goldman Sachs neither represents nor provides any assurances that the actual Reference Portfolio on the Closing Date or any future date will have the same characteristics as represented above. See the final Offering Circular for the Initial Reference Portfolio.

Reference Obligations are designated as "Midprime" herein if the weighted average FICO score of the underlying collateral that secures such Reference Obligation is greater than 625. All other Reference Obligations are designated as "Subprime" herein.

56

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)				
(2)				
(3)				
(4)				
(5)				

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool		
(2)		8% HPA for life		
(3)	Base	HPA slows to 5% by end-2005		
(4)				
(5)				

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool		
(2)		8% HPA for life		
(3)	Base	HPA slows to 5% by end-2005		
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter		
(5)	Meltdown	-5% for the next 3 years, 5% thereafter		

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool	1.4%	
(2)		8% HPA for life	3.2%	
(3)	Base	HPA slows to 5% by end-2005	5.6%	
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter		
(5)	Meltdown	-5% for the next 3 years, 5% thereafter		

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool	1.4%	
(2)		8% HPA for life	3.2%	
(3)	Base	HPA slows to 5% by end-2005	5.6%	
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter	11.1%	
(5)	Meltdown	-5% for the next 3 years, 5% thereafter	17.1%	

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool	1.4%	
(2)		8% HPA for life	3.2%	
(3)	Base	HPA slows to 5% by end-2005	5.6%	
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter	11.1%	
(5)	Meltdown	-5% for the next 3 years, 5% thereafter	17.1%	

- Actual HPA: -10% annualized from Q4, 2005 to Q4, 2008

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool	1.4%	
(2)		8% HPA for life	3.2%	
(3)	Base	HPA slows to 5% by end-2005	5.6%	
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter	11.1%	
(5)	Meltdown	-5% for the next 3 years, 5% thereafter	17.1%	

- Actual HPA: -10% annualized from Q4, 2005 to Q4, 2008
- Forecast losses as of 2/2009 in 2006-1 ABX from JPM: 23.44% (assuming -30% HPA in 2009!)

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool	1.4%	15%
(2)		8% HPA for life	3.2%	15%
(3)	Base	HPA slows to 5% by end-2005	5.6%	50%
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter	11.1%	15%
(5)	Meltdown	-5% for the next 3 years, 5% thereafter	17.1%	5%

- Actual HPA: -10% annualized from Q4, 2005 to Q4, 2008
- Forecast losses as of 2/2009 in 2006-1 ABX from JPM: 23.44% (assuming -30% HPA in 2009!)

Fact 10: Insiders biggest losers

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns
  - Merrill Lynch

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns
  - Merrill Lynch
  - Lehman Brothers

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns
  - Merrill Lynch
  - Lehman Brothers
  - Citigroup

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns
  - Merrill Lynch
  - Lehman Brothers
  - Citigroup
  - AIG

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns
  - Merrill Lynch
  - Lehman Brothers
  - Citigroup
  - AIG
- Who was short?

Fact 10: Insiders biggest losers

- Goldman Sachs arranged a deal:
- Bet on whether a collection of subprime backed securities would default.
- Who was long on such securities?:Insiders
  - Bear Stearns
  - Merrill Lynch
  - Lehman Brothers
  - Citigroup
  - AIG
- Who was short?
  - Paulson and Company

Losses to Insiders

- Mortgage-Related Losses to Financial Institutions from the Subprime Crisis, as of June 18, 2008.

	Institution	Loss ($ billions)		Institution	Loss ($ billions)
1	Citigroup	42.9	11	Washington Mutual	9.1
2	UBS	38.2	12	Credit Agricole	8.3
3	Merrill Lynch	37.1	13	Lehman Brothers	8.2
4	HSBC	19.5	14	Deutsche Bank	7.6
5	IKB Deutsche	15.9	15	Wachovia	7.0
6	Royal Bank of Scotland	15.2	16	HBOS	7.0
7	Bank of America	15.1	17	Bayerische Landesbank	6.7
8	Morgan Stanley	14.1	18	Fortis	6.6
9	JPMorgan Chase	9.8	19	Canadian Imperial (CIBC)	6.5
10	Credit Suisse	9.6	20	Barclays	6.3

Fact 11: Outsiders biggest winners

Fact 11: Outsiders biggest winners

- Was Paulson & Co. an insider? No.

Fact 11: Outsiders biggest winners

- Was Paulson & Co. an insider? No.
- Paulson had no experience in real estate

Fact 11: Outsiders biggest winners

- Was Paulson & Co. an insider? No.
- Paulson had no experience in real estate

> *"Paulson was a merger-arb guy and suddenly he has strong views on housing and subprime," [a potential investor] recalls. "The largest mortgage guys, including Vranos at Ellington, one of the gods of the market, were far more positive on subprime." (Zuckerman loc. 2281)*

What was Paulson's strategy?

What was Paulson's strategy?

- Not based on "inside information" about the deals

What was Paulson's strategy?

- Not based on "inside information" about the deals
- Believed that the key do mortgage defaults was house prices:

What was Paulson's strategy?

- Not based on "inside information" about the deals
- Believed that the key do mortgage defaults was house prices:

> *Paulson and Pellegrini concluded that the only way their trades would work was if the U.S. real estate market had reached unsustainable levels and began to fall..."*

What was Paulson's strategy?

- Not based on "inside information" about the deals
- Believed that the key do mortgage defaults was house prices:

  > *Paulson and Pellegrini concluded that the only way their trades would work was if the U.S. real estate market had reached unsustainable levels and began to fall..."*

- Big insight – "Eureka" moment.

What was Paulson's strategy?

- Not based on "inside information" about the deals
- Believed that the key do mortgage defaults was house prices:

  Paulson and Pellegrini concluded that the only way their trades would work was if the U.S. real estate market had reached unsustainable levels and began to fall..."
- Big insight – "Eureka" moment.

  Housing prices had climbed a puny 1.4 percent annually between 1975 and 2000, after inflation was taken into consideration. But they had soared over 7 percent in the following five years, until 2005. The upshot: U.S. home prices would have to drop by almost 40 percent to return to their historic trend line. (Zuckerman, loc 1936).

  The chart [showing that house prices would fall 40 percent] was Paulson's Rosetta stone, the key to making sense of the entire housing market.

Fact 13: Borrowers were optimistic about house prices

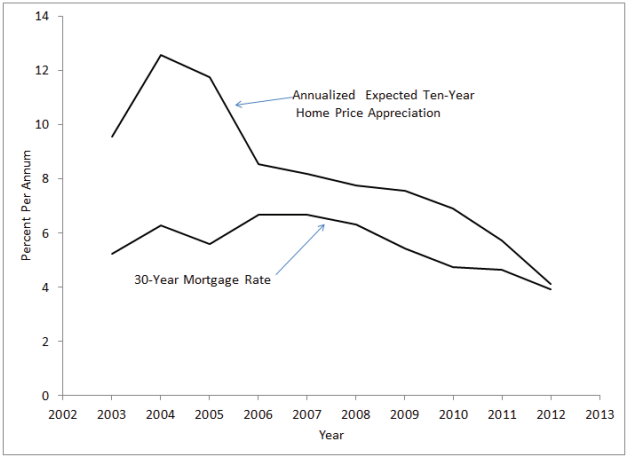
- "What have they been thinking? Home buyer behavior in hot and cold markets"

Fact 13: Borrowers were optimistic about house prices

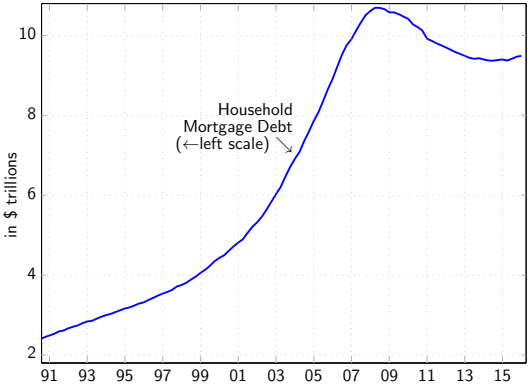
- "What have they been thinking? Home buyer behavior in hot and cold markets"
- *Brookings Papers on Economic Activity*, Fall 2012

Fact 13: Borrowers were optimistic about house prices

- "What have they been thinking? Home buyer behavior in hot and cold markets"
- *Brookings Papers on Economic Activity*, Fall 2012

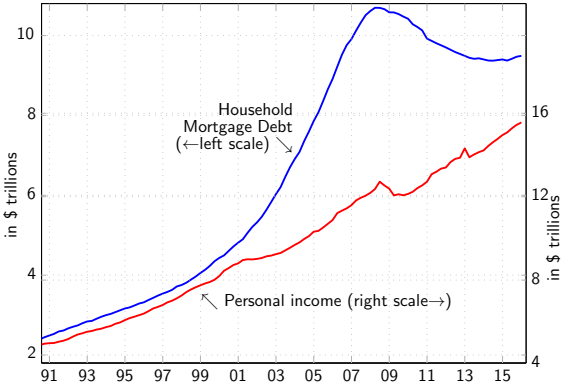


Debt Growth



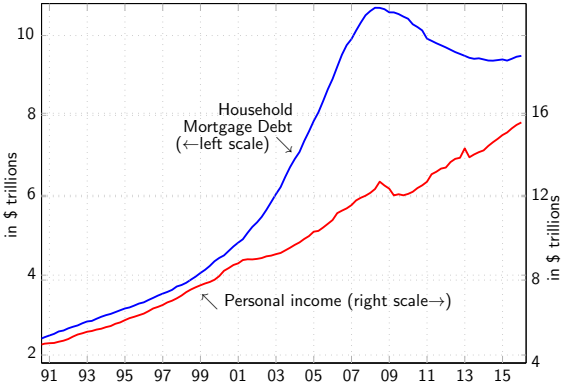
- Huge debt boom between 2001 and 2007

Debt Growth



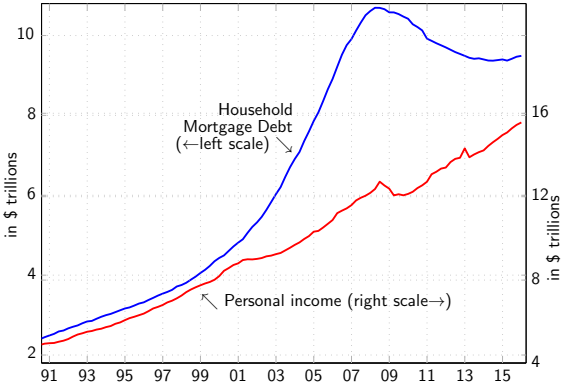
- Huge debt boom between 2001 and 2007
- Much faster than income

Debt Growth



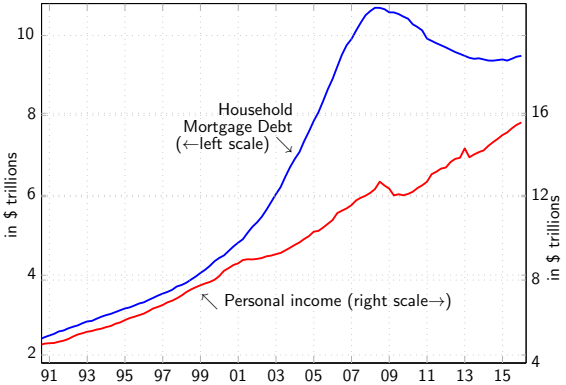
- Huge debt boom between 2001 and 2007
- Much faster than income

Debt Growth



- Huge debt boom between 2001 and 2007
- Much faster than income

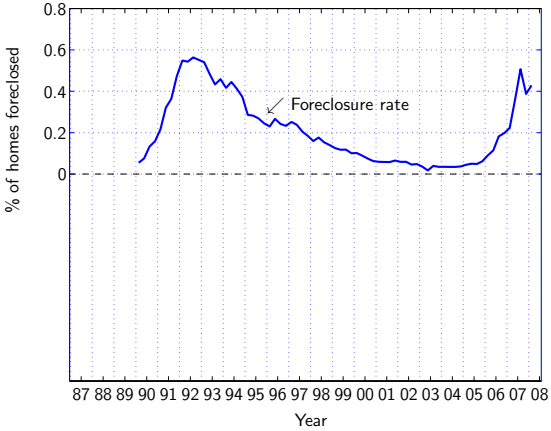
Debt Growth



- Huge debt boom between 2001 and 2007
- Much faster than income
- Questions:
  - Where did debt grow?
  - What was the role of subprime? Securitization?
  - What was the role of the government?

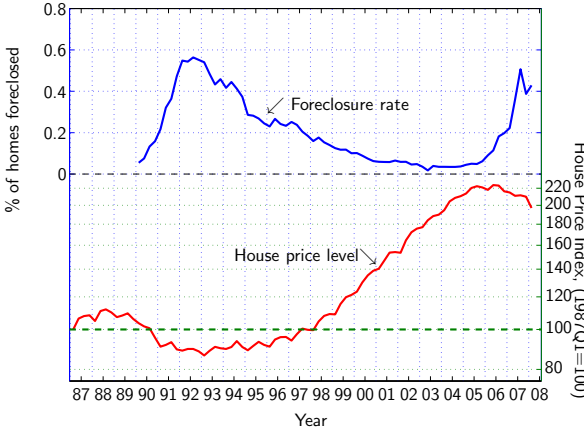
Mortgage Default (March 23)

- Why do people default on mortgages?
  - Negative equity
  - Role of income
  - "Strategic Default"
  - Bad underwriting and credit expansion



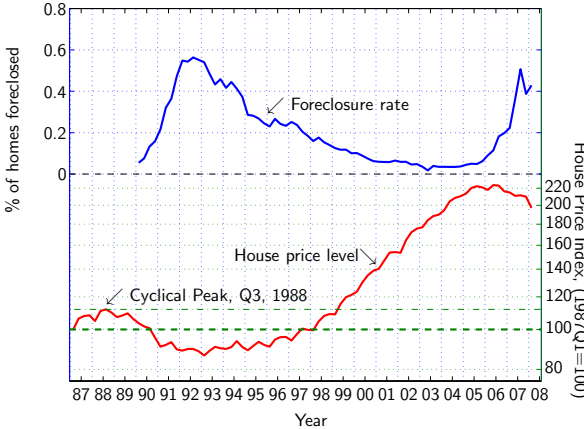
Mortgage Default (March 23)

- Why do people default on mortgages?
  - Negative equity
  - Role of income
  - "Strategic Default"
  - Bad underwriting and credit expansion



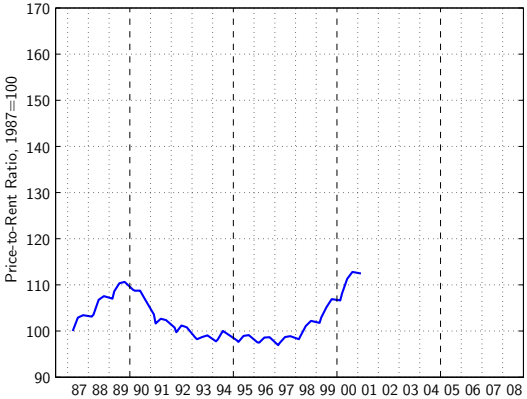
Mortgage Default (March 23)

- Why do people default on mortgages?
  - Negative equity
  - Role of income
  - "Strategic Default"
  - Bad underwriting and credit expansion



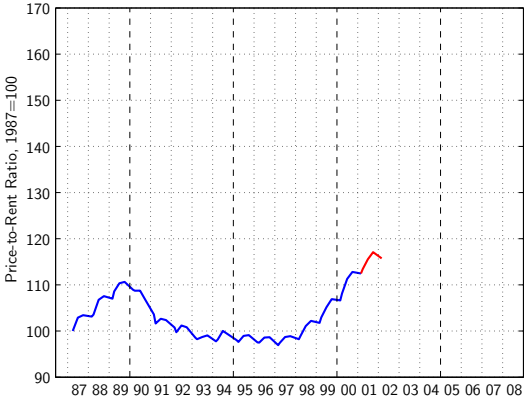
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs



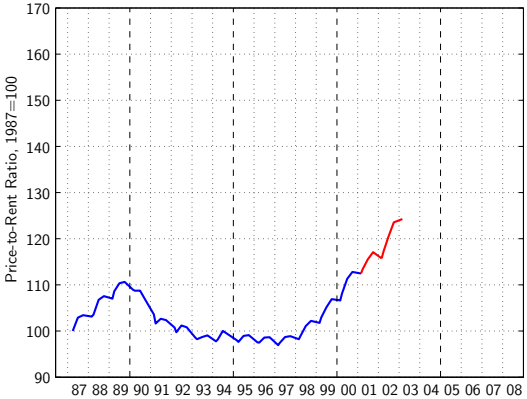
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs



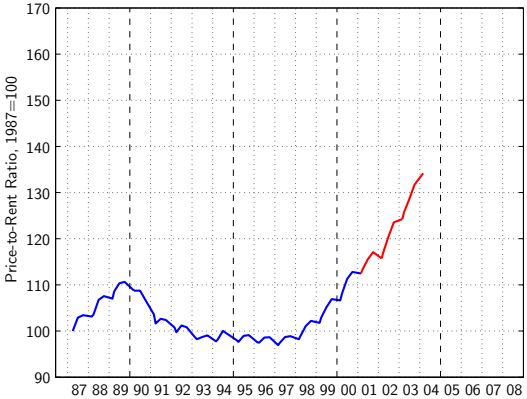
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs



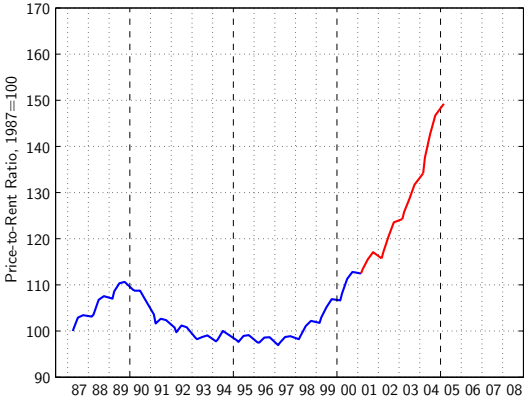
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs



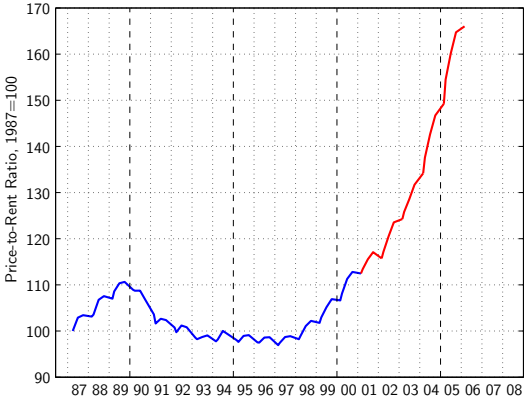
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs



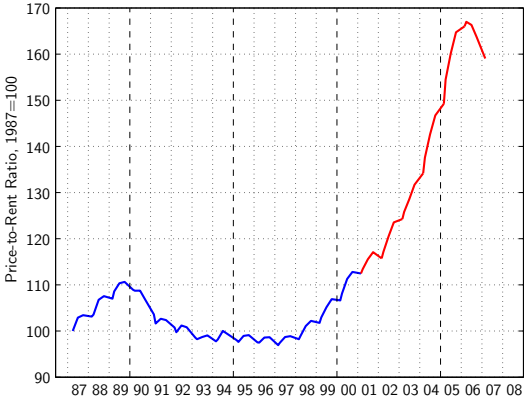
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs



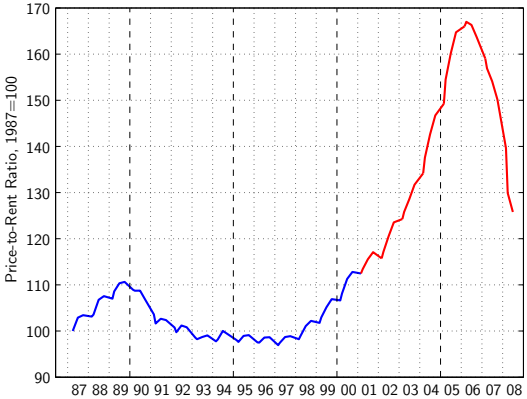
House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs

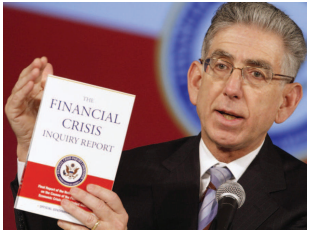


House prices (March 30)

- Why did house prices go up and down?
  - Financial innovation
  - Interest rates
  - Beliefs

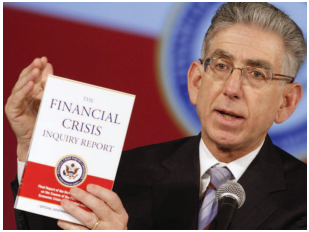


The Role of Institutions (April 6)



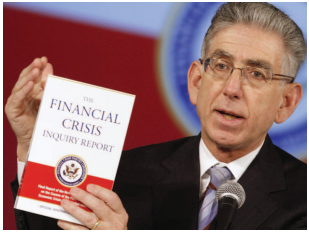
- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

The Role of Institutions (April 6)



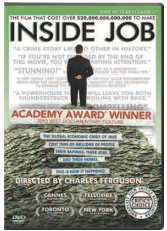
- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

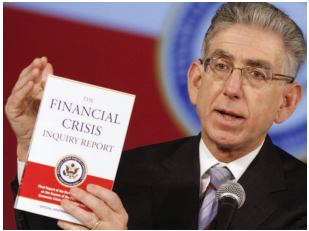
The Role of Institutions (April 6)



- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

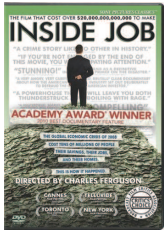
The Role of Institutions (April 6)

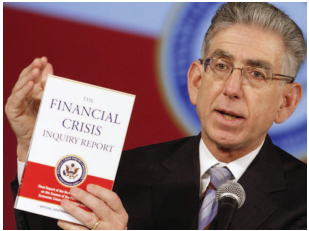




- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

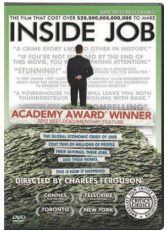
The Role of Institutions (April 6)

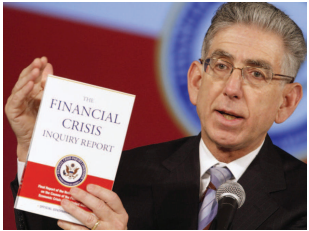




- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

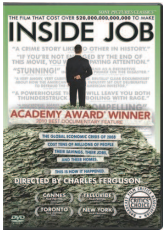
The Role of Institutions (April 6)

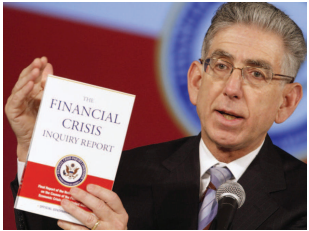




- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

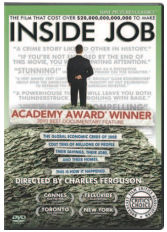
The Role of Institutions (April 6)

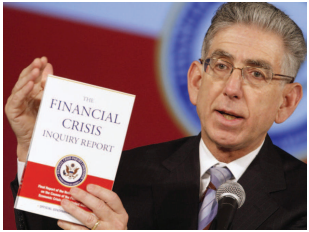


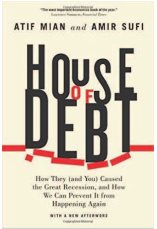


- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

The Role of Institutions (April 6)







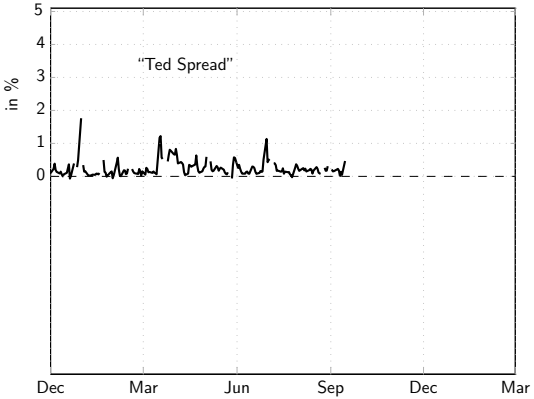
- Main narrative. Institutional problems
  - Securitization
  - Rating Agencies
  - Loan Servicers

Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses

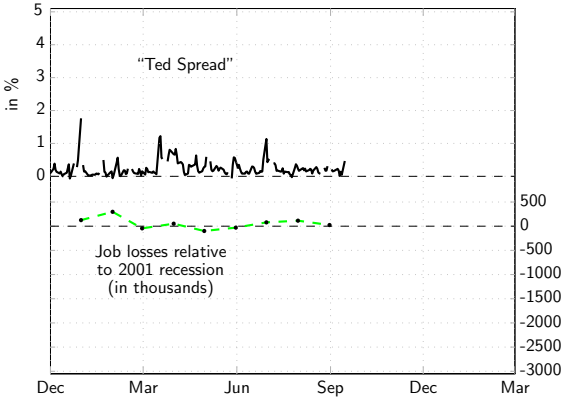
Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses



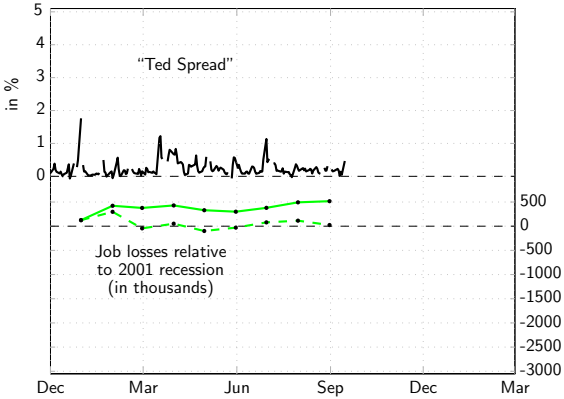
Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses



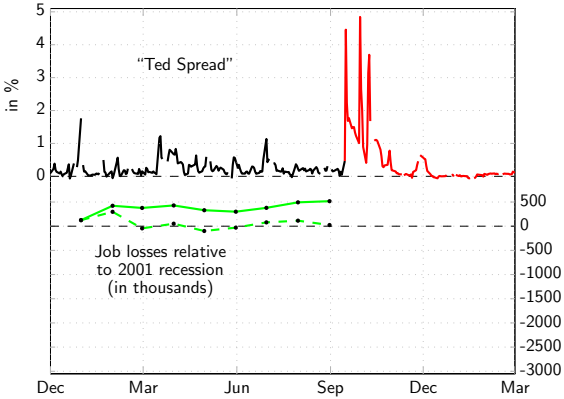
Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses



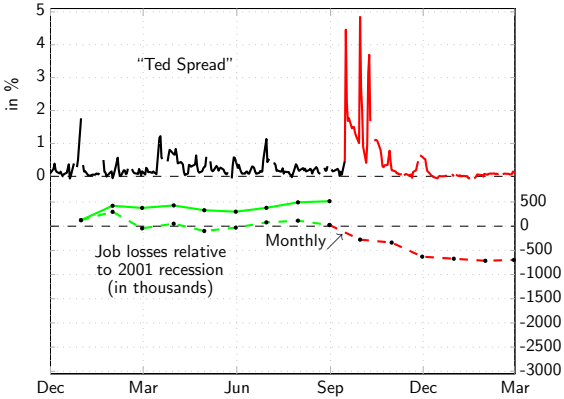
Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses



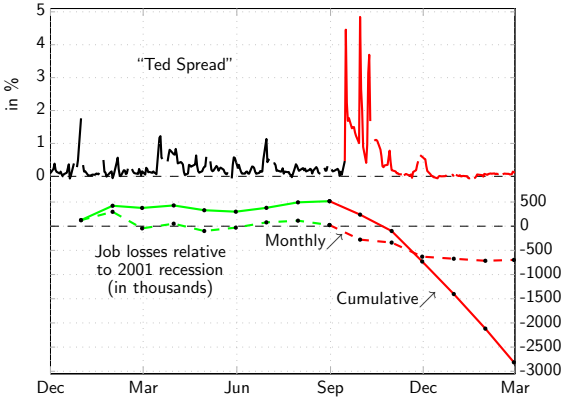
Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses



Financial Crisis (April 13)

- How did losses in mortgage markets trigger a financial panic
  - Measuring the panic
  - Role of derivatives and other financial products
  - Policy responses



Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)		
(2)		
(3)		
(4)		
(5)		

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)		
(5)		

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafte
(5)	Meltdown	-5% for the next 3 years, 5% thereafter

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafte
(5)	Meltdown	-5% for the next 3 years, 5% thereafter

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafte
(5)	Meltdown	-5% for the next 3 years, 5% thereafter

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafte
(5)	Meltdown	-5% for the next 3 years, 5% thereafter

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafte
(5)	Meltdown	-5% for the next 3 years, 5% thereafter

Expectations (April 20)

- Expectations are crucial to understanding crisis
  - Rational expectations
  - Measuring expectations
  - Modeling expectations

#	Name	Scenario
(1)	Aggressive	11% HPA over the life of the pool
(2)		8% HPA for life
(3)	Base	HPA slows to 5% by end-2005
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafte
(5)	Meltdown	-5% for the next 3 years, 5% thereafter

The slide you've all been waiting for...

The slide you've all been waiting for...

- The end.